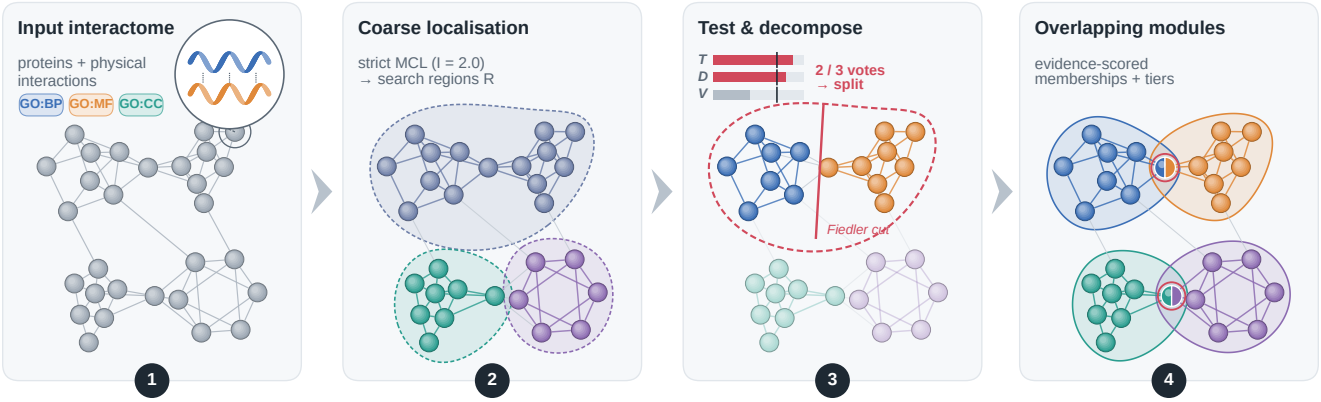
a Module construction

PPI graph and GO annotations only - no reference complexes



Membership evidence $S(p, C) =$ 0.45 τ topology + 0.35 σ semantic + 0.20 g GO terms

admit if $S \geq \theta_C$ (region $q_{0.25}$) and topology evidence guard passes

b Diagnostics exported with every run

- **Region record** — signals, thresholds and votes per region
- **Split audit** — objective terms of each alternative + reason
- **Membership ledger** — τ, σ, g, S, guard, tier and provenance
- **Perturbation stability** — 10% edge removal × 20, full reruns

Reference complexes never flow upward into a or b

c Evaluation only

reference complexes enter here

- **Reference** — Complex Portal, frozen + checksummed
- **Agreement** — 1:1 Jaccard ≥ 0.5; module P / R / F1
- **Overmerging** — regions holding ≥ 2 complexes
- **Tier support** — Core vs non-Core agreement
- **Uncertainty** — paired bootstrap; sensitivity grids